

MD. RAKIB MAHMUD HRIDYOY

Android Engineer · Mobile Application Developer

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[Github](#) · [Portfolio](#) · [LinkedIn](#)

PROFESSIONAL SUMMARY

Mobile engineer with 5+ years building and shipping production applications in Kotlin, Jetpack Compose, and Flutter. Experienced across real-time communication systems — SIP/VoIP telephony over Asterisk and self-hosted video conferencing infrastructure — and in delivering feature modules within Clean Architecture and BLoC codebases. Comfortable owning features end-to-end, from API integration through Play Store release.

TECHNICAL SKILLS

Native Android: Kotlin, Java, Jetpack Compose, Coroutines & Flow, Hilt, Room, Retrofit, Material Design, Google Play Billing, Android Profiler.

Architecture: MVVM, MVI, MVC, Clean Architecture, multi-module design, dependency injection, design patterns, data structures & algorithms, offline-first patterns.

Flutter: Dart, BLoC / flutter_bloc, Riverpod, GetIt, GoRouter, Drift, Dio, Firebase Crashlytics & Remote Config.

Other Cross-Platform: React Native, JavaScript, TypeScript.

Real-Time & Telephony: SIP, VoIP, Asterisk AML (asterisk-java), Jitsi stack (Prosody, Jicofo, JVB, Coturn), WebRTC.

Backend & Data: RESTful API design, MySQL, SQLite, ORMLite + SQLCipher, Supabase (Postgres), Firebase services (Auth, Push Notifications), JWT authentication.

Tooling & Practice: Git, Docker, Azure, Nginx, Jira, Agile/Scrum, Figma (design implementation), Play Store release management, AI-assisted development (Claude Code, Antigravity, Gemini).

PROFESSIONAL EXPERIENCE

Software Engineer II | Brotecs Technologies Ltd, Dhaka, Bangladesh

December 2020 – July 2026

- Architected and migrated the flagship Android codebase to a multi-module structure, cutting incremental build times and allowing the team to work in parallel without merge conflicts.
- Shipped and maintained 5 production Android applications through full Play Store release cycles, including Google Play Billing integration, staged rollouts, and post-release crash triage.
- Led adoption of Jetpack Compose, migrating screens from legacy XML views and standardising a shared design-system component library used by the whole mobile team.
- Owned client-facing requirements gathering for enterprise telephony clients, translating operational needs into technical specifications and delivery timelines.
- Collaborated with design, QA, product, and offshore engineering teams in an Agile/Scrum process, contributing to code review and release planning across time zones.
- Introduced agentic AI tooling (Claude Code, Antigravity) into the team's development workflow for large-scale refactoring and debugging.

SELECTED PROJECTS

Employee HR & Attendance Platform — Flutter, Dart, REST

- Built the attendance and time-tracking module end-to-end — clock in/out, live punch status, and month-by-month attendance history — for an HR platform.
- Implemented manager approval workflows including bulk approve/reject, rejection-reason capture, and status filtering for early-leave and extra-time requests.
- Delivered all features within an existing Clean Architecture and BLoC codebase, following per-feature data/domain/presentation layering with GetIt dependency injection and GoRouter navigation.

Drill-Down Region Map Package (open source) — Flutter, Dart, CustomPainter, GeoJSON

- Directed AI-assisted development of a standalone, dependency-free Flutter package for an interactive region map — tap a division to reveal its districts, tap a district to reveal points inside it — built generic against any GeoJSON rather than tied to one country.
- Reviewed and tested throughout rather than writing the implementation line by line; owned the technical decisions — WCAG-based label contrast over a naive lightness check, pole-of-inaccessibility label placement for concave coastlines and multi-part shapes, and a hue-spacing color scheme proven collision-free for any number of regions.
- Ships with real UN OCHA administrative boundary data, simplified from over 200,000 polygon vertices to roughly 7,000 for smooth zoom animation. Covered by 500+ lines of tests. Now consumed as a live dependency in Chinhito, a Flutter travel app currently in development (see below). github.com/hridoyRakibMahmud/geo_drilldown.

Chinhito — Travel Discovery App (in development) — *Flutter, Riverpod, Supabase (Postgres), Google OAuth*

- Directed AI-assisted development of a Flutter app built on top of my own `geo_drilldown` package — a map-first destination explorer with login-optional browsing, live Supabase/Postgres data, and Google OAuth for account-gated features.
- Feature set built so far: a drill-down map explore screen backed by live database rows, a visited-places tracker tied to the map, a social feed with posting and likes, and a profile/stats screen. Reviewed and tested throughout; not yet published.

Cabin Crew Calling Console — *Android, Java, Asterisk AMI, SIP, OrmLite + SQLCipher*

- Replaced an aircraft's built-in hardware interphone console with an Android application, giving cabin crew the same call answering, forwarding, and cabin-extension routing on standard devices.
- Integrated asterisk-java to subscribe to the Asterisk AMI event stream and reconstruct live state for every extension and trunk — in-call, multi-call, and trunk usage — keeping the model in sync as events arrived so operators saw current status rather than stale data.
- Built the entire application interface, with call records and configuration stored locally in OrmLite behind SQLCipher encryption.

VoIP Calling Applications — *Android, Java, SIP, in-house SDK, OrmLite + SQLCipher, Firebase*

- Built softphone applications for organisations running their own telephony servers — internal extension-to-extension calling with outside calls routed through a configured trunk — deployable against any SIP server.
- Owned integration of the company's in-house communication SDK into the call-handling flow, ensuring placing, receiving, and routing behaved consistently in every direction.
- Deployed in aviation, carrying aircraft-to-ground, ground-to-aircraft, and aircraft-to-aircraft calling alongside internal cabin extensions.

Virtual Second-Number Application — *Android, Java, Kotlin, SIP, REST, Google Play Billing, Firebase Auth*

- Shipped a Google Play consumer application selling SIP-based virtual phone numbers, letting users hold an additional international number alongside their existing SIM and call internationally from it.
- Integrated external provider APIs through the cloud backend to source and manage number inventory, making numbers browsable and purchasable in-app.
- Implemented payments and subscriptions with Google Play Billing, handling failed payments and subscription state changes rather than assuming successful purchase.
- Implemented user authentication with Firebase Auth — phone number verification at launch, later extended to email-based sign-in through Google and Apple.

Enterprise Video Conferencing Platform — *Jitsi, Prosody, Jicofo, JVB, Coturn, Debian, Nginx*

- Delivered a licensed, organisation-based conferencing platform with a CMS for administrators to manage members, invite external participants, and schedule sessions.
- Took over the server deployment after an initial Ubuntu-based attempt stalled, provisioning and configuring the full Jitsi stack on Debian behind an Nginx reverse proxy until the system was operational.
- Hardened the deployment for production: JWT authentication, Prosody domain matching, Jicofo trusted-domain configuration, secure WebSocket upgrade, and firewall rules.

Telemedicine Consultation Application — *React Native, REST*

- Built the interface and API layer for a clinical application working with Mintti Vision, Heartbook ECG, and digital stethoscope devices — including consultation booking against recorded measurements. Internal SaaS product; scope was UI and API integration.

EDUCATION

B.Sc. in Computer Science and Engineering

Bangladesh University of Business and Technology (BUBT), Dhaka · February 2020 · CGPA 3.82 / 4.00

CERTIFICATIONS & TRAINING

- **Flutter Development Bootcamp with Dart** — Udemy, 2026
- **Mobile Application Development (Android)** — BITM, BASIS Institute of Technology and Management, 3 months, 2019
- **UI/UX Design** — ICT Ministry of Bangladesh, 3 months, 2018